Advay Mishra

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Education

New York University: Center of Data Science & Courant Institute of Mathematical Sciences — **New York, NY**

Bachelor of Arts in Joint Major of Computer Science and Data Science — *May 2025*

Relevant Coursework: Fundamentals of Machine Learning, Advanced Topics: Machine Learning for Climate Change, Artificial Intelligence, Natural Language Processing, Basic Algorithms, Predictive Analytics, Causal Inference, Data Management and Analysis

Experiences

Incoming Data Science Intern — **Remote**

IEEE Computer Society Student Branch Chapter (CS SBC), IIT Kharagpur — ***August 2025 - current***

Machine Learning Undergraduate Intern (Curricular Practical Training) — **New York, NY / Remote**

The Cornell Lab of Ornithology, Cornell University *(Mentors: Dr. Holger Klinck, Larissa Sugai)* | *Internship Report* — *May 2024 - Current*

- Conducting in-depth analysis of frog vocalization data in New England, focusing on climate-driven shifts in mating call timing.
- Applying advanced data science and machine learning techniques to examine bioacoustic patterns under expert guidance, enhancing understanding of amphibian behavioral responses to environmental changes.
- Identifying frog call regions of interest in spectrograms through machine learning, isolating specific vocalizations for analysis
- Preprocessing audio signals by removing background noise with segmentation and median clipping, and distinguishing high-quality signals through unsupervised clustering algorithms to maintain dataset integrity
- Using isolated frog calls to build a robust training dataset for a neural network, enabling species identification through unique vocalization patterns

Co-Founder & Undergraduate Research Assistant — **New York, NY**

The Venn Institute, New York University *(Mentors: Dr. Pascal Wallisch)* | *The Venn Institute* — *November 2023 - Current*

- Co-founded the Venn Institute, a think tank for interdisciplinary research, under the mentorship of Dr. Pascal Wallisch, focusing on data science applications to contemporary issues.
- Leading a project on Chinese dynasties, utilizing "TimeGraph" methodology to visualize territorial shifts over time.
- Conducting timeline video pre-processing, pixel extraction, and temporal validation to accurately track dynastic changes.
- Applying scale factor normalization and HSV-based color segmentation for consistent representation across timelines, contributing to a working paper on predictive patterns in empire dynamics.

Machine Learning Intern — **Remote**

Avolta Inc. | *Recommendation Letter* — *September 2023 - December 2023*

- Contributed to a user authentication system for automotive security with 90% success in testing.
- Developed and trained Elliptic Envelope, One-Class SVM and Isolation Forest models for user authentication via pressure sensor data, enhancing accuracy with dimensionality reduction.
- Implemented real time-time facial recognition and liveness detection with YOLOv8, OpenCV etc. to prevent spoofing.

Data Science Volunteer — **Remote**

FynCom — *August 2023 - October 2023*

- Scanned websites for fraud terms and modeled FTC complaints to forecast trends for firms like Apple and Microsoft.

Select Publications

- Analyzing Chinese dynasties using TimeGraph visualization and machine learning (working paper with Dr. Wallisch).
- Preserving Avifaunal Diversity in Santragachi Jheel, Howrah, West Bengal, India Published by the Asiatic Society, December 2020

Projects

Retrieval-Augmentation Generation Model

Final Project*, Natural Language Processing Course (2024)* | Project Link

- Developed a Retrieval-Augmented Generation (RAG) system leveraging NLP techniques to extract actionable insights from the CORD-19 dataset, processing over 43,000 scientific articles.
- Achieved 94.4% accuracy in generating concise, context aware answers, to COVID-19 related queries, showcasing expertise in information retrieval and natural language selection.

Song Genre Prediction Model

Capstone Project, *Fundamentals of Machine Learning Course (2023)* | Project Link

- Developed a model to classify song genres from 50,000 Spotify tracks using audio features.
- Applied T-SNE for dimensionality reduction and optimized clusters with k-means.
- Built a random forest classifier, achieving an AUC-ROC value of 0.99 and evaluated model reliability with an ROC curve.

Skills

Python, C, Assembly, SQL, TensorFlow, PyTorch, Machine Learning (Supervised & Unsupervised), Data Analysis & Regression, Object-Oriented Programming, Algorithm Development, Scikit-Learn